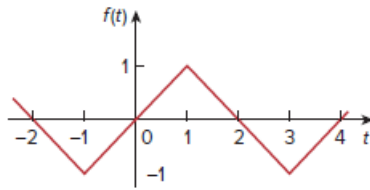


## Problem

Obtain the complex Fourier series expansion of  $f(t)$  in Fig. 17.17. Show the amplitude and phase spectra.

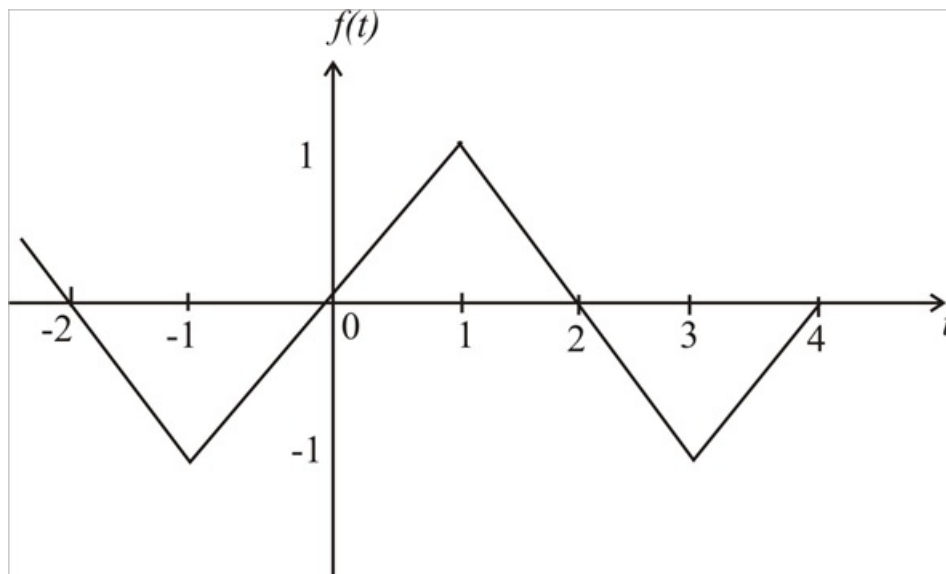
**Figure 17.17:**



## Step-by-step solution

## Step 1 of 7

Given wave form



## Step 2 of 7

$$f(t) = t \quad -1 < t < 1$$

Since  $T = 2$

$$\omega_0 = \frac{2\pi}{T}$$

$$\omega_0 = \frac{2\pi}{2}$$

$$\omega_0 = \pi \text{ rad/sec}$$

## Step 3 of 7

$$C_0 = \frac{1}{T} \int_{-\frac{T}{2}}^{\frac{T}{2}} f(t) dt$$

$$C_0 = \frac{1}{2} \int_{-1}^1 t dt$$

$$C_0 = \frac{1}{2} \left[ \frac{t^2}{2} \right]_{-1}^1$$

$$C_0 = \frac{1}{4} [1 - (-1)^2]$$

$$\therefore C_0 = 0$$

#### Step 4 of 7

$$C_n = \frac{1}{T} \int_{-\frac{T}{2}}^{\frac{T}{2}} f(t) e^{-jn\omega_0 t} dt$$

$$C_n = \frac{1}{2} \int_{-1}^1 t e^{-jn\pi t} dt$$

$$C_n = \frac{1}{2} \left[ \frac{te^{-jn\pi t}}{-jn\pi} \right]_{-1}^1 - \int_{-1}^1 \frac{e^{-jn\pi t}}{-jn\pi} dt$$

$$C_n = \frac{1}{2} \left[ \frac{e^{-jn\pi} - (-1)e^{jn\pi}}{-jn\pi} - \frac{e^{-jn\pi t}}{(-jn\pi)^2} \right]_{-1}^1$$

$$C_n = \frac{1}{2} \left[ \frac{e^{-jn\pi} + e^{jn\pi}}{-jn\pi} - \frac{e^{-jn\pi} - e^{jn\pi}}{(-jn\pi)^2} \right]$$

$$C_n = \frac{2\cos(n\pi)}{2(-jn\pi)} + \frac{2j\sin(n\pi)}{2(-jn\pi)^2}$$

$$C_n = \frac{(-1)^n}{-jn\pi}$$

$$\therefore C_n = \frac{j(-1)^n}{n\pi}$$

#### Step 5 of 7

Therefore, the complex fouries series expansion of function is,

$$f(t) = \sum_{\substack{n=-\infty \\ n \neq 0}}^{\infty} \frac{j(-1)^n}{n\pi} e^{jn\pi t}$$

The magnitude and phase of complex fouries series

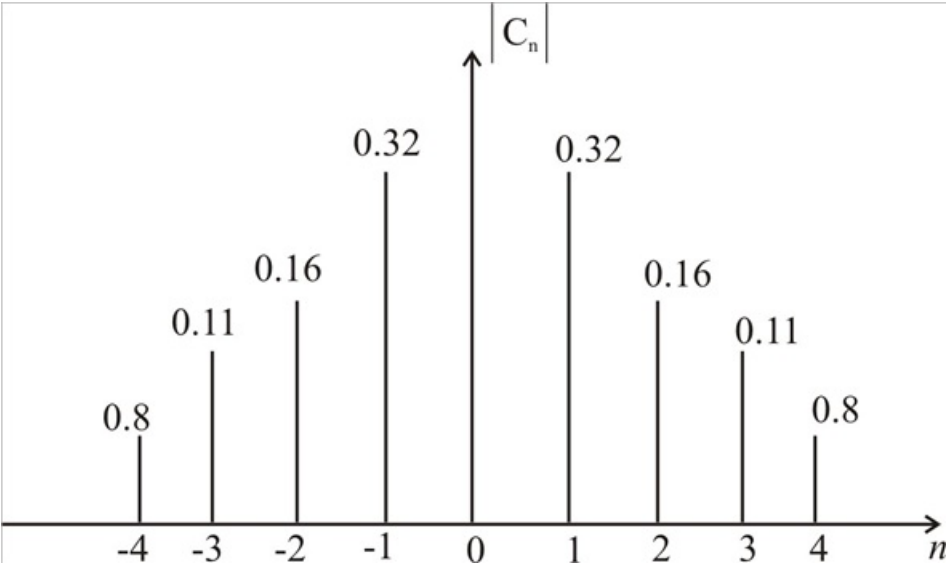
$$|C_n| = \frac{1}{n\pi} \quad n = -\infty, -2, -1, 0, 1, 2, \infty$$

$$\theta_n = -\tan^{-1}(\infty)$$

$$\theta_n = \begin{cases} -90^\circ & n \text{ is odd} \\ 90^\circ & n \text{ is even} \end{cases}$$

#### Step 6 of 7

The amplitude spectra,



Step 7 of 7

The phase spectra,

